

## ST PAUL DOWNTOWN, MN, USA

WMO: 726584

Lat: 44.932N

Lon: 93.056W

Elev: 700

StdP: 14.33

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14927

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -9.7       | -5.8 | -18.2                         | 2.1 | -8.4 | -14.8 | 2.5 | -4.9 | 26.1                  | 14.2 | 23.8 | 13.9 | 8.1                   | 310  | 0.558 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month<br>DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD<br>to 0.4% DB |      |
|---------------|---------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|-------------------------|------|
|               |                           | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                         |      |
|               |                           | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                    | PCWD |
| (a)           | (b)                       | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                     | (p)  |
| 7             | 17.8                      | 90.4            | 73.7 | 87.7 | 72.5 | 84.2 | 70.6 | 77.2                | 86.4 | 75.1 | 83.7 | 73.1 | 81.5 | 11.7                    | 180  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 74.1                            | 130.6 | 83.3 | 72.3 | 122.5 | 81.1 | 70.3 | 114.3 | 78.8 | 41.1          | 86.3 | 39.0 | 84.4 | 37.1 | 81.7 | 85.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |      |                    | n-Year Return Period Values of Extreme Temperature |           |      |            |      |            |       |            |       |
|-------------------|------|------|----------------------------|-------|------|--------------------|----------------------------------------------------|-----------|------|------------|------|------------|-------|------------|-------|
| 1%                |      | 2.5% | 5%                         | Mean  |      | Standard Deviation |                                                    | n=5 years |      | n=10 years |      | n=20 years |       | n=50 years |       |
| Min               |      | Max  |                            | Min   | Max  |                    | Min                                                | Max       | Min  | Max        | Min  | Max        | Min   | Max        |       |
| (a)               | (b)  | (c)  |                            | (e)   | (f)  | (g)                | (h)                                                | (i)       | (j)  | (k)        | (l)  | (m)        | (n)   | (o)        | (p)   |
| 23.4              | 20.2 | 18.3 | DB                         | -16.0 | 95.7 | 6.1                | 3.1                                                | -20.4     | 98.0 | -24.0      | 99.8 | -27.4      | 101.6 | -31.8      | 103.9 |
|                   |      |      | WB                         | -16.6 | 80.6 | 5.9                | 2.3                                                | -20.8     | 82.2 | -24.2      | 83.6 | -27.5      | 84.9  | -31.8      | 86.6  |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 46.9   | 16.2  | 20.5  | 33.1  | 47.1  | 59.2  | 69.0  | 73.8  | 71.3  | 63.7  | 49.8  | 35.2  | 22.0  |
|                                                                                  | DBStd       | 22.20  | 12.32 | 12.09 | 11.93 | 9.94  | 8.42  | 6.47  | 5.48  | 5.31  | 8.18  | 8.98  | 10.36 | 11.39 |
|                                                                                  | HDD50       | 4035   | 1047  | 825   | 538   | 168   | 17    | 0     | 0     | 0     | 5     | 114   | 453   | 868   |
|                                                                                  | HDD65       | 7381   | 1512  | 1245  | 990   | 540   | 220   | 35    | 5     | 9     | 121   | 478   | 893   | 1333  |
|                                                                                  | CDD50       | 2900   | 0     | 0     | 14    | 82    | 302   | 569   | 739   | 661   | 415   | 108   | 10    | 0     |
|                                                                                  | CDD65       | 768    | 0     | 0     | 1     | 3     | 39    | 154   | 278   | 205   | 81    | 7     | 0     | 0     |
|                                                                                  | CDH74       | 6327   | 0     | 0     | 5     | 49    | 404   | 1265  | 2431  | 1504  | 604   | 65    | 0     | 0     |
|                                                                                  | CDH80       | 1967   | 0     | 0     | 0     | 12    | 122   | 391   | 837   | 426   | 168   | 11    | 0     | 0     |
| Wind                                                                             | WSAvg       | 8.3    | 8.8   | 8.9   | 8.9   | 9.5   | 8.6   | 7.7   | 7.3   | 6.9   | 7.8   | 8.5   | 8.7   | 8.4   |
| Precipitation                                                                    | PrecAvg     | 33.1   | 0.8   | 0.7   | 1.7   | 3.0   | 4.2   | 4.9   | 4.5   | 4.6   | 3.3   | 2.7   | 1.7   | 1.1   |
|                                                                                  | PrecMax     | 40.9   | 2.3   | 1.5   | 4.1   | 6.3   | 8.7   | 9.3   | 11.2  | 8.7   | 6.7   | 5.6   | 5.2   | 2.9   |
|                                                                                  | PrecMin     | 23.9   | 0.1   | 0.1   | 0.4   | 0.9   | 0.9   | 2.2   | 1.9   | 1.1   | 0.6   | 0.7   | 0.1   | 0.2   |
|                                                                                  | PrecStd     | 4.4    | 0.5   | 0.4   | 0.9   | 1.3   | 1.9   | 1.9   | 2.1   | 2.0   | 1.9   | 1.5   | 1.4   | 0.6   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 45.0  | 52.3  | 70.6  | 80.9  | 89.9  | 93.0  | 95.7  | 92.4  | 89.9  | 81.3  | 66.4  | 50.2  |
|                                                                                  |             | MCWB   | 38.4  | 45.7  | 57.9  | 61.8  | 68.7  | 74.5  | 77.3  | 74.6  | 72.0  | 65.9  | 55.2  | 45.1  |
|                                                                                  | 2%          | DB     | 38.7  | 45.3  | 62.3  | 72.7  | 83.0  | 88.3  | 90.6  | 88.0  | 84.3  | 73.5  | 59.9  | 44.3  |
|                                                                                  |             | MCWB   | 35.0  | 40.9  | 53.2  | 56.1  | 65.2  | 71.9  | 75.4  | 73.4  | 70.5  | 61.3  | 51.4  | 39.7  |
|                                                                                  | 5%          | DB     | 35.9  | 40.1  | 55.2  | 68.0  | 78.8  | 84.1  | 87.8  | 84.2  | 81.1  | 68.9  | 54.5  | 39.1  |
|                                                                                  |             | MCWB   | 33.4  | 36.0  | 47.8  | 53.2  | 63.0  | 69.7  | 73.5  | 71.4  | 68.3  | 58.8  | 47.4  | 35.9  |
|                                                                                  | 10%         | DB     | 33.2  | 36.6  | 49.5  | 63.4  | 73.3  | 81.2  | 84.3  | 81.7  | 76.9  | 63.6  | 49.8  | 36.4  |
|                                                                                  |             | MCWB   | 31.0  | 33.4  | 43.4  | 50.6  | 59.8  | 67.1  | 71.4  | 69.5  | 66.4  | 55.6  | 44.2  | 33.7  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 38.5  | 46.0  | 60.5  | 63.5  | 72.8  | 77.6  | 80.8  | 79.2  | 75.7  | 69.3  | 58.3  | 47.3  |
|                                                                                  |             | MCDB   | 43.3  | 50.7  | 67.1  | 78.1  | 84.5  | 88.7  | 89.3  | 88.6  | 84.0  | 76.0  | 64.3  | 48.5  |
|                                                                                  | 2%          | WB     | 35.5  | 40.7  | 54.2  | 58.5  | 67.8  | 74.7  | 78.0  | 75.9  | 72.7  | 63.9  | 52.9  | 39.7  |
|                                                                                  |             | MCDB   | 37.8  | 45.0  | 61.0  | 69.0  | 78.8  | 85.0  | 87.3  | 84.0  | 81.1  | 70.3  | 58.9  | 43.4  |
|                                                                                  | 5%          | WB     | 33.1  | 36.8  | 48.4  | 55.0  | 65.2  | 71.9  | 75.8  | 74.0  | 70.5  | 60.0  | 48.0  | 36.5  |
|                                                                                  |             | MCDB   | 35.6  | 40.1  | 55.1  | 65.7  | 74.6  | 81.3  | 84.3  | 81.4  | 78.0  | 66.9  | 53.3  | 39.2  |
|                                                                                  | 10%         | WB     | 30.8  | 33.5  | 43.4  | 52.0  | 62.4  | 69.4  | 73.6  | 71.8  | 68.0  | 56.5  | 44.4  | 33.9  |
|                                                                                  |             | MCDB   | 32.8  | 36.4  | 48.7  | 61.9  | 71.3  | 77.7  | 82.3  | 79.1  | 75.0  | 63.2  | 49.4  | 36.2  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 14.4  | 15.2  | 16.3  | 18.6  | 19.2  | 18.0  | 17.8  | 17.5  | 18.3  | 16.6  | 14.3  | 12.8  |
|                                                                                  |             | MCDBR  | 16.5  | 17.5  | 24.0  | 26.3  | 25.3  | 22.4  | 20.7  | 20.1  | 22.0  | 24.6  | 21.1  | 16.8  |
|                                                                                  |             | MCWBR  | 13.4  | 13.4  | 15.6  | 14.4  | 11.9  | 10.2  | 9.2   | 9.2   | 10.6  | 13.7  | 13.7  | 13.5  |
|                                                                                  | 5% WB       | MCDBR  | 15.6  | 16.6  | 22.8  | 23.5  | 22.0  | 19.7  | 18.8  | 17.9  | 19.1  | 20.3  | 19.5  | 15.5  |
|                                                                                  |             | MCWBR  | 13.0  | 13.0  | 15.5  | 14.4  | 12.1  | 10.8  | 10.2  | 10.2  | 11.0  | 13.3  | 13.7  | 12.8  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.270  | 0.289 | 0.325 | 0.381 | 0.412 | 0.404 | 0.401 | 0.408 | 0.379 | 0.326 | 0.294 | 0.271 |       |
|                                                                                  | taud        | 2.415  | 2.343 | 2.368 | 2.284 | 2.251 | 2.319 | 2.348 | 2.332 | 2.377 | 2.518 | 2.511 | 2.435 |       |
|                                                                                  | Ebn at Noon | 272    | 285   | 288   | 279   | 272   | 275   | 274   | 267   | 266   | 270   | 261   | 259   |       |
|                                                                                  | Edh at Noon | 23     | 30    | 33    | 40    | 42    | 40    | 38    | 37    | 33    | 25    | 21    | 20    |       |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 547    | 891   | 1206  | 1457  | 1689  | 1880  | 1993  | 1664  | 1298  | 824   | 526   | 384   |       |
|                                                                                  | RadStd      | 63     | 91    | 110   | 148   | 137   | 148   | 71    | 118   | 91    | 110   | 49    | 43    |       |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (57 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page